Reg. No. | | | | | | | | | | |

[12ME54/13ME54]

FIFTH SEMESTER B.E/B.TECH EXAMINATION, DECEMBER-2015

DYNAMICS OF MACHINES

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

1. A four bar mechanism under the action of two external forces is shown in figure 1. Determine the torque to be applied on the link AB for static equilibrium of the mechanism. **14**

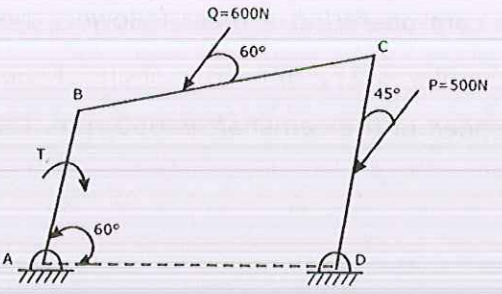

Fig. 1

AB = 50mm
BC = 66mm
CD=55mm
CE=25mm
CF=30mm
AD=100mm
$B\hat{A}D = 60°$

2. **a.** State and explain D'Alembert's principle. **08**

b. What are the requirements of an equivalent dynamical system? **06**

3. The turning moment diagram for a Four-Stroke gas engine may be assumed for simplicity to be represented by 4 triangles, the areas of which from the line of zero pressure are as follows: Sanction Stroke = $0.45\times10^{-3}\ m^2$ **14**

Compression Stroke = $1.7\times10^{-3}\ m^2$

Expansion Stroke = $6.8\times10^{-3}\ m^2$

Exhaust Stroke = $0.65\times10^{-3}\ m^2$

Each m^2 of area represents 3MN-m of energy. Assuming the rising torque to be uniform, find the mass of the rim of a flywheel required to keep the speed between 202 and 198 rpm. The mean radius of the rim is 1.2m.

4. Four masses m_1, m_2, m_3 and m_4 are 200kg, 300kg, 240kg and 260kg respectively. The corresponding radii of rotation are 0.2m, 0.15m, 0.25m and 0.3m respectively and the angles between successive masses are 45°, 75° and 135°. Find the position and magnitude of the balance mass required, if its radius of rotation is 0.2m. Use graphical method. **14**

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5. A V-twin engine has the cylinder axes at right angles and the connecting rods operate a common crank. The reciprocating mass per cylinder is 11.5 kg and the crank radius is 75mm, length of connecting rod is 0.3m. Show that the engine may be balanced for primary force by means of a revolving mass. Also find the maximum secondary force if the engine speed is 500 rpm. **14**

6. a. Write a short note on gyroscope. **04**

b. An aeroplane makes a complete half circle of 50 metres radius, towards left, when flying at 200km per hour. The rotary engine and the propeller of the plane has a mass of 400kg and a radius of gyration of 0.3m. The engine rotates at 2400 rpm clockwise when viewed from the rear. Find the gyroscopic couple on the aircraft and state its effect on it. **10**

7. In a symmetrical tangent cam operating a roller follower, the least radius of the cam is 30mm and roller radius is 17.5mm. The angle of ascent is 75° and the total lift is 17.5mm. The speed of the camshaft is 600 rpm. Calculate the principal dimensions of the cam. **14**

8. a. Define: **02**

i. sensitiveness and

ii. stability of governors

b. A porter governor has equal arms each 250 mm long and pivoted on the axis of rotation. Each ball has a mass of 5kg and the mass of the central load on the sleeve is 15kg. The radius of rotation of the ball is 150 mm when the governor begins to lift and 200mm when the governor is at maximum speed. Find the minimum and maximum speeds and range of speed of the governor. **12**